Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/20/2013
PI Name:	Stutte, Gary Ph.D.		
Project Title:	Symbiotic Nodulation in a Reduced Gravity Environment (SyNRGE)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	321-861-3493
Organization Name:	QinetiQ-North America/NASA Kennedy Space Center		
PI Address 1:	ESC-Team QNA		
PI Address 2:	Space Life Sciences Laboratory		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899	**Congressional District:**	24
Comments:	NOTE: PI had a change of employment from Dynamac Corporation to QinetiQ-North America, effective 1 March, 2011.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	03/01/2011	**End Date:**	04/30/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135 ; ISS NOTE: New end date is 4/30/2014, per NSSC information (Ed., 2/19/14) NOTE: New end date is 2/28/2014, per NSSC information (Ed., 8/6/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roberts, Michael (CASIS-Center for the Advancement of Science in Space)		
Grant/Contract No.:	NNX11AL06A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	SyNRGE (Symbiotic Nodulation in a Reduced Gravity Environment) will investigate microgravity effects associated with microbe-host interactions and cell-cell communication using a plant-bacteria model system. Seeds from a single cultivar of the model legume species Medicago truncatula (barrel medic) will be germinated on-orbit in the presence of genetically marked strains of nitrogen-fixing bacteria of the species Sinorhizobium meliloti capable of forming a mutualistic endosymbiosis with leguminous plants. Cellular, molecular, and morphological analyses of plant tissue specimens and bacteria fixed at serial time intervals on-orbit will be compared to ground controls to elucidate the effect of microgravity on plant and bacterial physiological responses, mechanisms of cell-cell communication and resource exchange (i.e., carbon and nitrogen), and the extent of symbiotic nodule formation. Symbiotic nitrogen-fixing bacteria and leguminous plants have evolved complex signal exchange mechanisms that enable specific bacterial species to induce specific host plant species to form invasion structures through which the bacterium enters the plant root. Once inside, the bacteria are enclosed within a micro-aerobic cell compartment by the host plant and differentiate into specialized cells that express an oxygen-sensitive enzyme to catalyze the conversion of nitrogen to ammonia for use by the host plant. Although the rhizobia-legume microbe-host relationship is mutually beneficial, rhizobial infections of plants can be deleterious. Rhizobia are similar to Brucella spp. that infect animals (including humans) in that they both form chronic infections of eukaryotic cells within a host-derived membrane compartment and require host-derived factors for survival within the host. Based on recent reports of increased virulence of Salmonella typhimurium, Pseudomonas aeruginosa, and Streptococcus pneumoniae in reduced gravity, differences in the infectious and/or pathogenic responses of rhizobial bacteria or the systemic acquired resistance of the host plant are also likely. [Ed. note: 1/19/2012] The Principal Investigator (Gary Stutte, PhD) and co-Investigator (Mike Roberts, PhD) had a change of employment from Dynamac Corporation, prime contractor of the Life Science Services Contract (LSSC), at Kennedy Space Center to QinetiQ-North America, the prime contractor for the Engineering Services Contract (ESC) at Kennedy Space Center effective 1 March, 2011.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The SyNRGE experiment investigates the role of microgravity in development of biological nitrogen fixation in legume species. This symbiosis between a legume and a nitrogen fixing bacteria is directly and indirectly responsible for over 20% of the protein in human diets. SyNRGE results are investigating underlying mechanisms of this symbiosis that has wide ranging agricultural and environmental applications.
Task Progress:	The SyNRGE experiment designed to determine if the nitrogen fixing bacteria, S. meliloti, infect M. truncatula and initiate physiological changes associated with nodule formation. Roots of five-day-old M. truncatula cultivar Jemalong A17 (Enod11::gus) seedlings were inoculated 24 hr before launch with either S. meliloti strain 1021 or strain ABS7 and integrated into BRIC-PDFU hardware before being placed in a 4°C Cold Bag for launch on Atlantis. Inoculated plants and uninoculated controls were maintained in the dark at ambient temperature in the middeck of STS-135 for 11 days before fixation in RNAlaterTM by crew activation of the PDFU. These samples STS-135 crew preserved these samples on orbit with RNAlaterTM so that the state of the plants and bacteria in orbit could be seen after their return to Earth. Results from SyNRGE demonstrated good growth of the M. truncatula on orbit, with the control seedlings showing significantly longer roots (14.8 mm vs. 10 mm) in the flight set over the ground controls. RNA was extracted from flight and ground control samples fixed in RNAlaterTM and microarray analysis was performed using the GeneChipÒ Medicago Genome Array (Affymetrix) to determine whether changes in the activation of genes were associated with the µg environment or the combination of host plant and symbiotic S. meliloti utilized. The array contains over 61,200 probe sets, and the initial analysis revealed significant changes in associated with exposure to microgravity. The number of genes up-regulated was significantly reduced in the presence of the S. meliloti lines. This data is being summarized and prepared for publication. Initial assessment of the effect of microgravity exposure on the ability to form nodules indicated that significant cultivar differences existed. The super-nodulating line SUNN, which is an ethylene mutant, failed to thrive after return to 1 g, and less than 25% of the plants produced nodules. This was in sharp contrast to the ground controls, in which over 50% of the roots formed nodules. Image analysis data has been obtained from over 2600 images of individual shoots, and growth rate, and timing root growth, appearance of nodules, and nodule number determined. This data is being summarized for final publication.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Stutte GW, Roberts MS. "Microgravity Effects on the Early Events of Biological Nitrogen Fixation in Medicago Truncatula: Results from the SyNRGE Experiment." ESA/ISGP – Life in Space for Life on Earth Symposium, Aberdeen, United Kingdom, June 18-22, 2012. ESA/ISGP – Life in Space for Life on Earth Symposium, Aberdeen, United Kingdom, June 18-22, 2012. Abstract KSC-2012-116. http://naca.larc.nasa.gov/search.jsp?R=20120010667&qs=N%3D4294932186%2B4294817689%26Nn%3D4294937145%257CDocument%2BType%257CJournal%2BArticle , Jun-2012
Papers from Meeting Proceedings	Stutte GW, Roberts MS. "Nitrogen Fixation in Medicago truncatula: Results from the SyNERGE experiment." ESA/ISGP – Life in Space for Life on Earth Symposium, Aberdeen, United Kingdom, June 18-22, 2012. In: Proceedings of Life in Space for Life on Earth : 18-22 June 2012, Aberdeen, United Kingdom. (editor: L. Ouwehand). ESA Special Publication 706. Noordwijk : ESA (European Space Agency), 2013. ISBN 978-92-9221-270-4. CD Rom. , Feb-2013
Significant Media Coverage	Herridge L. "Research Summary, 'Atlantis' Final Mission Included Successful Kennedy-Developed Plant Experiment' " NASA Press Release. Posted on NASA Mission Pages and submitted as NASA Press Release 8/1/2011.http://www.nasa.gov/mission_pages/shuttle/behindscenes/bricsynrge.html , Aug-2011
Significant Media Coverage	Russel Q. (Mona Lisa Productions) "Documentary Film 'Power of Plants' had significant feature of SyNRGE impacts and Landing activites. Documentary first aired in France, won special prize in European Film Festival, and will be show on National Geographic, USA." Documentary Film, 'Power of Plants, 'Mona Lisa Productions, July 2012., Jul-2012